Assignment - 06

Q1.
→ A user-defined function is a function created by the programmer using def keyword.
1. To avoid writing same code repeatedly.
2. Improve code reusability.
3. Reduce program size.

Q2.
→ Syntax : def f-name (parameters) :
function () body.
statements.

The colon (:) tells python that the func() body begins after this line. PVM uses this colon along with Indentation to identify the block of code.

Q3.
→ Python uses indentation to define the block of code that belongs to a function.
·] Identifies code block / func() body.
·] Improves readability.

Q4.
→ A] function Defination.
·) Creates the function
·) Uses def keyword.
·) Executed only once while defining.

B] Function Call.

- Executes the function.
- Can be executed multiple times.

eg- def hello ():
        print ("Hello!")   } → defination.

hello ()   ⟶ function call.

Q5.

→ A function can exists with no body using pass statement.

eg. def Add () :
        pass

When we write pass, it acts as a placeholder until the actual code is written.

Q6.

→ The pass statement is a null statement. It does nothing when executed.

It is used:

- As a placeholder for future code.
- To avoid syntax error when a function body is temporarily empty.

Q7.

→ Parameters → Variable listed in function def^n.

eg- def add (a , b) ⟶ Parameters.

Arguements → Actual values passed to the func() during the func() call.

eg- add (10, 20) → Arguements.

Q8.

→ We get TypeError → missing / extra arguement

Q9.

→ A return value is the result sent back to the caller using the return statement.
Once the return statement is executed, the func() ends & the returned value can be stored or used.

Q10.

→ YES! Python function can return multiple values.

eg-
```
def calculate (a, b) :
        return ((a+b), (a-b))

x, y = calculate (10, 5)
print (x)
print (y)
```

Output → 15 (Adds)
5 (Subtracts)